



Assessing the quality of services at an academic library

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ABSTRACT

Customers usually have high expectations on the services they receive. The LibQUAL model was employed in this study to investigate the quality of services at an academic library. The participants were chosen from the five colleges in a university using simple random sampling. Two hundred participants were chosen from each college. In all, 1000 participants (including faculty members, postgraduate and undergraduate students) were selected for the study. The study revealed that users knew and used the library's services. The researchers delivered the questionnaires to the respondents at the library. The results showed that the library service quality (LSQ) fell short of users' expectations. There was a statistically significant difference in LSQ between gender of users. The users' expectations of the library's level of service were out of sync or mismatch. The variations could be attributed to differences in information needs of users. This requires the library to assess its services from users' perspective regularly. The use of the LibQUAL model provides useful information that library management can employ for developing service quality measurement scale and planning for service quality. The study also provides information about services that needs improvement, so that library staff could manage users' expectations and satisfaction in a better way. University authorities expect good return of investment made into the development of the library. Therefore, libraries should improve upon their services to boost the image of the Universities.

1. Introduction

The use of resources in academic libraries can have an impact on users' expectations. The academic library serves as a central location for information [1,2]. Academic libraries strive to serve its users or clients with high-quality services. The users of academic libraries include faculty members, researchers, teaching and non-teaching staffs, students and community members. The quality of library services offered to clients or users may have a favorable impact on their information needs and seeking behavior.

In recent years, research on quality of service has become an important growing area for librarians and information professionals [1,3–6]. Determining the type of services offered and the best strategies to enhance service delivery are both aided by evaluating or assessing the services. Assessing the services aids in identifying areas where resources might be improved [3]. The process of reviewing service quality helps to identify what is effective and should be kept up with, as well as what changes should be made [4].

To provide quality service, academic libraries must assess their infrastructural facilities and information resources. Access to

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current learning resources, learning commons rooms and up-to-date modern facilities are some of the expectations of users [5]. However, these have become a challenge. Some university libraries are confronted with issues such as power outages, expensive fuel costs, and insufficient access to e-databases [6,7].

In Africa, lack of funds to run academic libraries affects quality of service delivery [5,8]. According to Refs. [5,8], librarians need to deal with the services they provided to achieve quality service delivery despite lack of funds. As a result, it is critical for university or academic libraries to improve their service delivery, because the desire for information and knowledge is the driving force behind learning. The manner in which librarians serve their users determines the level of service.

In the view of [1], quality of services provided to users ensures that customers are satisfied. Quality of service means that the academic librarians or information professionals find the best way to provide what users want [1,9]. Librarians are responsible for assisting library clients or users in obtaining the information they need. In the view of [9], academic libraries' major goal has been to meet the demands of its patrons or users. Academic library users' expectations, information seeking behavior, and information demands are only a few of the elements driving libraries to enhance the services they offer to information users.

The researchers' interests have been inspired by self-observation and the concerns raised by some clients about poor library service delivery. This observation supports the view of [10], who indicated that organizational culture, and employee training affect service delivery. Some users or clients complain about poor services given to them at the front desks or the circulation desks and some of the staff are not able to provide them with the services they needed. Additionally, library users or clients continue to have increased expectations about materials or resources available in and through the library and how to get special services. Library users are more casual about library use and the type of services library staff offers them. Clients usually have high expectations on the service quality provided by libraries. Service quality is a major factor that drives decision of clients to prefer a particular library [5]. Academic libraries offer almost same variety of services, but their quality may differ. Also, service quality is one of the main determinants of clients' satisfaction. In the view of [11], when service quality is inadequate, it may be underused by clients. The researchers wanted to investigate the quality of services offered at an academic library. Which qualities of library's service satisfy or fall short of clients or users' expectations. Furthermore, improving the understanding on gender differences if any on service quality is beneficial for information professionals. Therefore, it is also necessary to investigate the quality of services at an academic library based on gender, if there is, so that library management can assess the level of service quality and identify gap for improvements. These are research gaps that needs to be filled. In this study, the LibQUAL model was used to investigate the library services. The study is guided by the following research questions.

Research question 1. What are the services users are aware and used in an academic library?

Research question 2. Which qualities of academic library's service satisfy or fall short of clients or users' expectations?

Research question 3. Is there any significant difference in perceived quality of services at an academic library based on gender?

2. Literature review

Examining the quality of service is one of the key ways academic libraries use to achieve their goals and aims [12]. Assessing the services provided to library users will shape the goals and aims of academic libraries. Assessment of library resources for example physical space, available collections, and internet connectivity help to ascertain the progress of work [13].

There are various models that can be used to evaluate service quality. For instance, the lunservice performance-only model, higher education performance model, and SERVQUAL model can be adopted to assess service quality. Because it was relevant, the LibQUAL + model was used.

2.1. LibQUAL+™ model

The Association of Research Libraries (ARL) developed LibQUAL+ in cooperation with numerous Texas A&M University professors, including Colleen Cook [8]. The model developed allowed libraries to discover their strengths and weaknesses [14]. According to Ref. [15], the LibQUAL + model helps to identify disparities between an expected service and what customers actually receives from service providers.

Although [16], revealed that LibQUAL + model lacks interpretability when seeking to judge overall library success, the researchers hoped that the model can still be used as a measurement tool to predict user's intention in using the library. The LibQUAL + model gives opportunity for managers of on establishment to appreciate the concerns of clients or service users [1,6,9,13]. Similarly, according to the [17], the LibQUAL+ is a well-known and accepted model that libraries use to measure, interpret, and act on users' opinions about service delivery. This approach has been frequently used in library studies to assess service quality [1,3]. The LibQUAL + model focuses on users' opinions and demands to improve on libraries services.

LibQUAL measures three elements of service quality: affect of service (AS), information control (IC) and library as place (LP) [18]. The AS dimension relates to the perceptions of users about library staff competency, abilities and support [18]. The AS dimension deals with how the client or user is being treated in the library. Similarly [19], explains in a similar manner that the physical facilities, equipment, employees, and communication materials' look are among the LP variables. The LP took into account areas that encourage study, quiet and isolation, a place for introspection, and a setting conducive to contemplation.

The library's collection is investigated by the IC dimensions. It assesses whether the collections are adequate to satisfy the demands of library users. One of the main goals of the library is to make sure that a full collection of various resources has been gathered for

users to access and use. It is crucial that library personnel assist patrons in finding material that satisfies their demands for knowledge and educate them on how to do so effectively.

2.2. Quality of service

[1] define quality of service as services provided to ensure that clients or customers are satisfied. This indicates that the user's impression of the service defines and measures the quality of the service supplied [12]. Therefore, quality must conform to standards. According to Ref. [20], if suitable information sources are not available for users to use, the quality of service suffers. By this way, the expectation of services provided may influence user's expectations.

2.3. Users' satisfaction of quality of service

Users' satisfaction is an assessment of whether a service meets users or clients' expectations [21]. The library offers its clients or users information resources. Because a library cannot function without its patrons, it is vital that libraries perform an assessment to know users' level of satisfaction. Users' satisfaction of quality of service can be linked to services received, delivering effectiveness, appropriateness and positive standard of delivering. The importance of including library users' perspectives in the development and improvement of service delivery quality cannot be overstated. Academic libraries should allow their users to evaluate, judge, and determine the quality of their services. This evaluation aids in appreciating user's expectations.

2.4. Expectations of users

Expectations are desired wants [9]. Users' expectations can only be met, according to Ref. [21], if library administration is notified of service deficiencies. Typically, expectation projects user's wishes, prompting them to continue using library services. User perceptions should, in an ideal scenario, surpass anticipated expectations. The "gap" is the discrepancy between what users expect and what they receive [22]. This gap might be beneficial or harmful. A positive gap indicates that customers have attained or obtained the actual expected level of service while a negative gap shows that service delivery falls short of what users had anticipated [23]. User expectations are important in this study because if they are not met, it may affect the user's assessment on quality of service.

3. Methodology

This quantitative study looked at the service quality at the library using a survey research methodology. The researchers used a cross-sectional approach. The cross-sectional technique was adopted since it made data collection easier. The cross-sectional design allows for the comparison of multiple variables at the same time. Five colleges of University of Cape Coast (UCC) were divided into five strata. The participants were chosen from the various Colleges using simple random sampling. About 10,000 participants made up of each College. In total, 200 participants were chosen from each college. A simple random sampling technique was applied to draw a sample of 5 colleges.

The participants from the 5 colleges were taken through convenient sampling due to the unavailability of complete lists of the library users. Only those participants were included in the sample who had experience of the library use and were either permanent faculty members, undergraduate or postgraduate students in the Colleges. Various types of users such as faculty members, undergraduate and postgraduate students can have different opinions on LSQ. There were 200 participants from each College, having different genders (undergraduate's $n = 410$, postgraduate student's $n = 390$, and $n = 200$ faculty members). The total sample size was set at $n = 1000$ (200×5), which is a reasonable sample size. In all, 1000 participants (including 200 faculty members, 390 postgraduates and 410 undergraduate students) were selected for the study. Only those with prior library experience were chosen to participate. Therefore, 10 % participants were randomly sampled from each College.

[24] collection instrument on LibQUAL was adopted for this study. The LibQUAL approach was chosen because it aids in identifying gaps between the services that a customer expects to receive and those that they really do. In the view of [15], LibQUAL + model allow information professionals to comprehend how customers view and assess their services. Some of the questions in the questionnaire were restructured to suit the respondents. The researchers asked questions on the services provided and the usage off the library, the extent infrastructure enhance service quality and how to meet users' expectations. Users' expectations for this enhanced version on LSQ were thought to be simple to understand. The AS, IC, and LP dimensions made up the 22 statements that made up the LibQUAL. However, after modifications, 19 statements were used in this study. IC statements were 6, AS 8 and 5 LP statements. The researchers delivered the questionnaires to the respondents at the library. Utilising Cronbach's alpha statistics, a reliability analysis test was conducted to evaluate the questionnaire. The Cronbach's alpha coefficient was 0.81, indicating that the instrument was reliable. Cronbach's alpha subscales for AS, IC, and LP, respectively, looked to be 0.83, 0.79, and 0.78 when further investigated. These values are acceptable using LibQUAL own definition compared to Ref. [25] measure of service quality.

The researchers adhered to the ethical principles of objectivity, anonymity and confidentiality. The researchers received ethical approval from the UCC Institutional Review Board (UCCIRB) before data collection. The ethical approval number is UCC/IRB/A/2020/2355. A period of two months was used for data collection. This helped the researchers to convince most of the participants to participate in the study. A written consent letter was obtained from all the participants. The data analysis was conducted applying descriptive and inferential statistics (independent sample *t*-test, ANOVA) with the help of Statistical Package for Social Sciences (SPSS), version 21.0.

4. Results

The results section presents data on gender. The researchers wanted to find out if there is any gender difference in LSQ so that information professionals can assess the level of service quality and identify gaps for improvements.

Table 1 show that females account for 43.5 %, while males account for 56.5 %. The male preponderance was a true depiction of the university population in most African countries.

4.1. Awareness and usage of library services

Table 2 shows users' awareness and usage of the services offered by the university.

According to the findings, most of the postgraduate students are aware of and used the reference service, lending, electronic resources and periodicals. Furthermore, the results also showed that the outreach and user education services were underutilised in terms of awareness and usage.

In terms of the undergraduates, most of them were aware of and used the reference service, photocopying, lending, electronic resources and periodical services. The library's outreach services were the least known among participants.

According to data on faculty members' knowledge or awareness of library services, the majority of them were aware of and used the photocopying services, lending, electronic resource and periodicals. According to the responses, outreach service and user education are not known by most users. The fact that a sizeable number of the users were not aware of some services is worth noting.

4.2. Attributes of library service quality

The attributes were considered using means and standard deviation. The data in Table 3 reveals that most of the LSQ variables do not meet their expectation.

There was a wide range of replies from participants in terms of the aspects of perceived and desired LSQ. In terms of negative gap, the following three services had the biggest differences: "the library website support information searching"; "the library electronic support services are available any time and "the library has more electronic resources". The following LSQ characteristics, on the other hand, had a marginally smaller difference between the perceived and intended levels of service quality: "library staff has the knowledge to answer the users' questions", "the library has a place that encourage conducive learning environment", "library staff is always ready to respond to the users' questions", "library staff can motivate users" and "library staff are courteous".

4.3. Participants' differences in terms of dimension

According to the LSQ dimension-wise gap, user opinion differed on three key dimensions: AS, IC and LP. The AS dimension relates to the perceptions of users about library staff competency and abilities. The LP variable also takes into account the physical look of the facilities, equipment, and communication resources made available to users. The LP considered items that facilitate study, comfortable location, and contemplative environment. The library's collection is also investigated by the IC dimensions. It examines whether the collections are adequate to satisfy the demands of library users. The details are represented in Table 4.

There were negative LSQ gaps in all three dimensions in the responses of the respondents. Table 4 shows that the negative gap in the IC dimension was the largest (−0.88), followed by the AS (−0.78). The lowest gap was observed in the LP dimension (−0.61), indicating slightly better services as compared to the IC and AS. The dimension gap of the LSQ all fell short of the users' expectations. Although, in Table 2, participants claimed that they were aware of and utilized the library's services, so the researchers employed the ANOVA and the independent sample *t*-test to see if there were any significant differences between the responses in the user groups. The details are provided in Tables 5 and 6.

4.4. The quality of library services based on user group experience

The researchers wanted to see if there was a statistically significant difference between the different user groups. The specifics are shown in Table 5.

A statistically significant difference was found as ($F = 6.06$, $p = 0.000$) among faculty members, undergraduate and postgraduate students. The mean values of −1.10 showed that the highest gap in LSQ was in the faculty group (−1.10), followed by the postgraduate students (−0.72), and the undergraduate students (−0.70). As a result, participants such as faculty members and the two types of

Table 1
Gender distribution of the participants.

Category	Gender		Total
	Female	Male	
Faculty members	95	105	200 (20 %)
Postgraduates	168	222	390 (39 %)
Undergraduates	172	238	410 (41 %)
Total	565 (56.5 %)	435 (43.5 %)	1000

Table 2

Respondents view and usage of the library services.

	Postgraduates		Undergraduates		Faculty members	
Services	Awareness	Usage	Awareness	Usage	Awareness	Usage
Outreach	21.0	5.0	23.0	5.5	44.5	23.5
User education	57.0	39.0	62.5	41.5	45.0	33.0
Reference	75.0	67.0	81.5	68.0	88.0	76.5
Photocopying	77.0	55.0	60.0	50.0	79.0	71.0
Lending	79.0	50.0	71.5	21.0	90.0	64.0
Electronic/Internet Resources	85.0	67.0	75.6	58.0	86.5	64.0
Periodicals	87.0	51.0	69.0	51.0	86.0	63.0

Table 3

Attributes of participants' expectations based on LibQUAL.

Items	Mean	SD
The library has enough learning space for categories of students	2.48	1.98
The library is structured for me to search for information easily	2.24	1.97
Library staff show professionalism in performance of duties	3.22	1.45
The library has more current printed materials	2.70	1.43
The library staff support me in my information searching	2.65	1.43
The library has electronic and print resources	2.35	1.40
The library has more electronic resources	1.36	1.38
Library staff can respond to questions from users	3.24	1.31
The library has space for group discussions	3.48	1.30
The library staff understand the learning needs of each library user	3.19	1.30
The library staff deals with each user in an ethical manner	3.20	1.29
The library staff are courteous	3.49	1.28
Library staff are ready to help users	3.10	1.28
The library is located at a comfortable location	2.65	1.25
Library staff can motivate users	2.39	1.25
The library electronic support services are available any time	2.39	1.25
The library has conducive learning spaces for users	3.66	1.21
The library website support information searching	2.10	1.19
The library is a place that encourage conducive learning environment	3.50	1.10

Mean and standard deviation was used to describe the central tendencies and the level of dispersion respectively. On the scale of 1–5, the mean describes where the answers from the respondents were tilted towards while the standard deviation described the extent to which the answers deviate from the mean. A standard deviation of between 0 and 2 as seen from Table 3 shows that the data is normally distributed.

Table 4

Participants' differences in terms of dimensions.

Dimension	Perceived Mean	Desired Mean	Service Gap
IC	2.54	3.42	-.88
AS	3.20	3.98	-.78
LP	3.50	4.11	-.61
Overall	3.07	3.81	-.74

*The observed total variability within a collection of data is split into two halves using the statistical analysis technique known as analysis of variance (ANOVA).

* Sample *t*-test: This statistical hypothesis test is used to ascertain whether a population mean that is unknown is distinct from a given value.

Table 5

The quality of library services depending on the user groups.

User category	N	Perceived Mean	Desired Mean	Gap	SD
Faculty members	200	3.20	4.30	-1.10	1.02
Postgraduate students	390	3.36	4.08	-.72	.78
Undergraduate students	410	3.28	3.98	-.70	1.10
Total	1000	3.26	4.08	-.82	1.18

Table 6
ANOVA statistics for library service quality based on the user groups.

	Sum of Squares	Mean Square	F	Sig.
Faculty members	26.01	7.58	1.783	.191
Students	1390.21	1.36	6.06	.000
Total	1410.71			

students were discovered to have differing perspectives on LSQ (see detail in [Table 6](#)).

[Table 6](#) presents the statistical differences on library service quality among two groups. These groups were the faulty members on one side and students (both undergraduates and postgraduates) on the other side. It could be observed that the F statistic is 6.06 with a corresponding significance value of 0.000 which is lower than 0.05. The F value demonstrate a statistical difference while the significance value illustrate that the differences is significant. The implication, therefore is that quality of library services is perceived differently from the perspectives of faulty members and students. This might be due to their information needs and information search skills.

4.5. Gender difference in library service quality

The researchers were also curious in the gender difference in satisfaction with LSQ. The independent sample *t*-test was used to see whether there was a difference in satisfaction with LSQ between male and female participants. The findings demonstrated a statistically significant difference ($F = 2.54$, $p = 0.00$) between male and female ($F = 3.32$, $p = 0.36$) participants. Again, the female participants showed a higher negative gap (-1.0) than the male respondents (-0.73). However, male participants had a more positive attitude toward the LSQ than female participants. The details are provided in [Table 7](#).

5. Discussion

The present study revealed that users were aware of the academic library's services. This is proven by the participants' ability to select the exact library's services they knew from a pre populated list. According to the findings, the reference service is the service that most participants are aware of and use. This finding is contrary to the study by Ref. [16] which indicated that the service most students of Ashesi University College and Ghana Institute of Journalism know and use the most is the lending service.

Outreach services were the least known and used. It appears to be the case with all the categories of users. This could be due to the low publicity of these services to the University community. Also, competition with other service providers could account for the low patronage of the library user education service. Improving awareness of library services and resources should be a continuous intervention. It is imperative for library management to prioritize publicity of its services to all users using approaches such as leaflets, pull-ups, flyers and media platforms. The institution can strengthen their orientation periods to engage users on the services available at the library. The introduction of academic writing and teaching of information literacy can help clients or users appreciate the services available at the library. It's critical to establish a positive working connection with new clients or users. This will give them the opportunity to introduce the library's resources and the assistance available for them.

Although, the users were aware of most of the services available at the library, there is still more room for improvement such as the provision of adequate electronic resources. There is need for library management to prioritize student engagement to obtain regular feedbacks on user satisfaction to determine which area of the services need improvement. This is so important because achieving library quality of service improves the image of the library.

The study's findings revealed that the LSQ fell short of the clients' expectations. There was a negative gap between users' expectation scores and the actual scores. The greatest disparity was discovered in the IC, followed by the AS. The narrowest gap was discovered along the LP. On the other hand, services related to information searching and accessibility of electronic collections meet users' expectations. This conclusion supports the view of [22,26] who concluded that library services did not meet the desired expectation of users.

The attributes of a given library service quality differ among faculty members and students. The implication, therefore is that quality of library services is perceived differently from the perspectives of faculty members and students. This might be due to their information needs and information search skills. This supports the view of [27], who indicated that users' academic needs were not stable.

The findings of the independent sample *t*-test demonstrated a statistically significant difference in LSQ between gender of the

Table 7
Gender difference on service quality.

Gender	N	Perceived Mean	Desired Mean	Mean	SD	F	T	P-value
Female	435	3.10	4.10	-1.0	1.22	3.32	-4.00	.36
Male	565	3.31	4.04	-.73	1.17	2.54	-3.55	.000

*N: Participants.

SD: Standard deviation.

library users. Male participants showed a lower negative gap than female participants, according to the data. These results support the work of studies conducted by Refs. [3,5] which showed that there is a considerable difference in the LSQ user groups between gender. On the contrary, the results were found to be inconsistent with studies conducted by Refs. [1,24] that found no gender-based disparities in LSQ.

Furthermore, the staff's demeanor in interacting with the consumers may be below the desired levels for users. This view supports the findings of [28], who concluded that employee's demeanor and interactions can affect customer relationship. In similar vein, of [29], also indicated that staffing challenges include inadequate staff, shortage of skilled staff and lack of continued professional staff in serving clients well. Furthermore, the library personnel must be encouraged to show a readiness to assist library users or customers and to provide them with attention. Thus, brand management efforts can be implemented to let library clients or customers feel at home with their quality services. The key is to provide library staff with appropriate customer service training, allowing them to be strategic and customer friendly.

6. Implications of the study

The study's conclusions have consequences for theory and practice. This study used LibQUAL as a guiding framework for investigating quality of service. The use of the LibQUAL model provides useful information that library management can employ for developing service quality measurement scale and planning improvement of service quality. The study provides a source of data that can assist other researchers. It also provides empirical data for effective quality of service management practices and contributes to the scholarly debate in LSQ.

The study also provides information about services that needs improvement, so that library staff could manage users' expectations and satisfaction in a better way. University authorities expect good return of investment made into the development of the library. Therefore, libraries should improve upon their services to boost the image of the Universities.

7. Conclusion and recommendation

Most of the participants used the library and this implied that the library was considered as an important place that foster academic excellence. Most of the services were known and used except outreach. Also, the library environment and the facilities were also a major concern to users. They required a comfortable learning atmosphere as well as good computer workstations that were connected to the internet. In terms of the services given, the general assessment of the level of service provided by the library was positive. There is more room for improvement.

The use of the LibQUAL + model allowed the researchers to understand how library users perceived the services provided them. The findings revealed a distinction between the library's clients' expectations of service quality. The variations could be attributed to differences in information needs of library users or clients. This requires the library to assess its services and resources from users' perspective regularly. The academic libraries should often use the LibQUAL model approach or use more accepted quality framework using longitudinal analysis with planned budget resources. There is also the need for user engagement. This will help management to get feedback from the users to improve upon the existing services. There is the need for library management to establish good policies, taking into consideration user preferences and expectations. There is also the need for librarians to implement user friendly approaches for the improvement of quality of service.

The study demonstrated a statistically significant difference in LSQ between gender of the library users. Male participants showed a lower negative gap than female participants, according to the data. This is not encouraging and call for more education on the services provided by the library. Library professionals must understand both users' expectations (male and female) as well as their individual needs to be able to serve them better. As a result, libraries must deliver exceptional service quality if they are to keep their clients or users and enhance their standing as a service provider. It is recommended that academic libraries put in place measures for the improvement of quality services at academic libraries that would eventually improve the quality of education and contribute to national development.

8. Limitation of the study and suggestions for further studies

The study had certain geographical and participant selection methods constraints. Further studies can be carried out on the same topic to cover other demographic variables and academic libraries in other developing countries. Further studies could also consider some elements of triangulation in study approach, user focus groups and user experience methods to assess clients or users' satisfaction. This will also help delve deeper in knowing the differences in taught between the different users (gender) of the library. The study further found a statistical difference on how faculty members and students perceive library service quality. However, the study did not include which among the groups (faculty members and students) perceive the service to be more quality than the other.

Data availability statement

Data will be made available on request.

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CRediT authorship contribution statement

Kwaku Anhwere Barfi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - review & editing. **Seth Kobina Parbie:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing - original draft, Writing - review & editing. **Christopher Kwame Filson:** Investigation, Methodology, Validation, Visualization, Writing - original draft, Writing - review & editing. **Mathias Vondee Teye:** Data curation, Methodology, Validation, Writing - original draft, Writing - review & editing. **Kwame Kodua-Ntim:** Data curation, Formal analysis, Visualization, Writing - original draft, Writing - review & editing. **Eunice Ayensu:** Data curation, Investigation, Project administration, Supervision, Validation, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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